The power of index notation, as we will see in subsequent chapters, is that all results derived in two dimensions will have the **same form** in higher dimensions. The indices will simply run over more values.

Having introduced the dot product, we now revisit the concept of the gradient. The change in the temperature function $T(x, y)$, as we move an infinitesimal distance is (using index notation)

$$dT = \frac{\partial T}{\partial x_1} dx_1 + \frac{\partial T}{\partial x_2} dx_2 = \sum_i \frac{\partial T}{\partial x_i}\, dx_i \ . \tag{108}$$

Notice that this may be written as the dot product

$$dT = \vec{\nabla} T \cdot d\bar{l} \ , \tag{109}$$

where

$$\vec{\nabla} T = \sum_i \frac{\partial T}{\partial x_i} \hat{e}_i \ , \tag{110}$$

is the Gradient of T and

$$d\bar{l} = dx_1\, \hat{e}_1 + dx_2\, \hat{e}_2 = \sum_i dx_i\, \hat{e}_i \ , \tag{111}$$

is the *length element* in two dimensions. Equation (109) is the two-dimensional analogue of equation (3) from the previous chapter on a one-dimensional world.

2.4.1 Vector fields and Divergence

Unlike a scalar field/function which takes a value at every point (like the temperature of a floor), a *vector field* associates a vector to every point in space. An example we have already encountered is the position vector

$$\bar{r} = x_1\, \hat{e}_1 + x_2\, \hat{e}_2 = \sum_i x_i\, \hat{e}_i \ . \tag{112}$$

We sketch this in Figure 8. Clearly an infinite number of arrows are needed since there are an infinity of points in the plane.

Since the operator $\vec{\nabla}$ is itself 'vectorial' in nature, it is natural to ask whether we can act it on a vector using the dot product introduced in (102). Indeed, given an arbitrary vector (field) we find

$$\vec{\nabla} \cdot \vec{A} = \sum_i \hat{e}_i \frac{\partial}{\partial x_i} \cdot \sum_j A_j \hat{e}_j = \sum_i \sum_j \frac{\partial A_j}{\partial x_i} \left(\hat{e}_i \cdot \hat{e}_j\right) \ . \tag{113}$$

The derivative does not act on the unit vector since it is constant. Thus

$$\vec{\nabla} \cdot \vec{A} = \sum_i \sum_j \frac{\partial A_j}{\partial x_i}\, \delta_{ij} \ = \sum_i \frac{\partial A_i}{\partial x_i} \ , \tag{114}$$

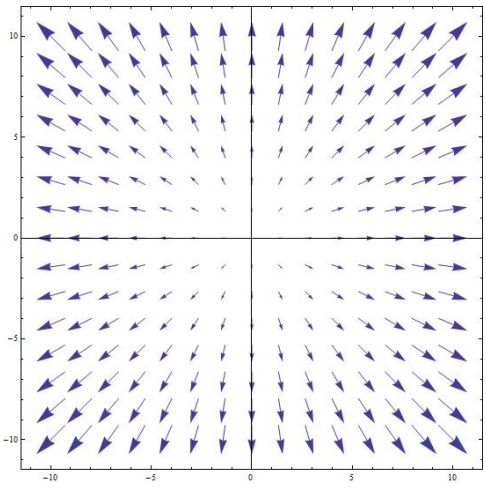


Figure 8: Position vector field

which defines the *divergence* of $\vec{A}$. It is important to note that the object $\vec{\nabla}$ wears two hats: it is a vector because its components transform correctly (allowing us to dot it with another vector) but it is also an operator since its components happen to be derivatives.

Example 9. Calculate the divergence of the position vector in (112)

From (114)

$$\vec{\nabla} \cdot \vec{r} = \sum_i \frac{\partial r_i}{\partial x_i} = \frac{\partial x_1}{\partial x_1} + \frac{\partial x_2}{\partial x_2} = 2 \ , \tag{115}$$

where r_i are the components of the position vector.

The divergence of a vector field is aptly named since it describes how much the vector 'spreads out' or diverges. In Figure 8, for example, the arrows clearly show a positive divergence (spreading out of the arrows as opposed to converging arrows which would happen if we considered $-\vec{r}$ instead).

If we were to imagine the vector field describing fluid flow then a positive divergence would indicate the presence of a source, like a tap for water.as in Figure 8. A negative divergence would then correspond to a sink which would have the opposite effect - representing water flowing down a drain or sink.

3 Three dimensions

3.1 Index notation

We now move to three spatial dimensions by introducing a new spatial dimension or direction, x_3, to enlarge the space spanned by the existing two directions x_1 and x_2. x_3 is chosen to be perpendicular to the plane spanned by the $x_1 - x_2$ axes. Correspondingly, we introduce a third unit vector so the *basis vectors* are now $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$. The key change, in moving from two to three dimensions, is that **all** indices $i, j...$ now run over the three values $1, 2, 3$.

As mentioned already, index notation guarantees that the form of all equations derived in two dimensions, will hold in three dimensions. Let us start for example with an arbitrary vector, in the space spanned by x_1, x_2 and x_3. This is written as

$$\vec{C} = \sum_i C_i \, \hat{e}_i \ , \tag{116}$$

identical in form to (96) **but** with the indices now running over three values. Explicitly written out, (116) reads

$$\vec{C} = \sum_i C_i \, \hat{e}_i \ \equiv \ C_1 \, \hat{e}_1 + C_2 \, \hat{e}_2 + C_3 \, \hat{e}_3 \ . \tag{117}$$

The index i, as before, is a dummy index since it is summed over. Given a second arbitrary *three-vector* (meaning a vector in three dimensions)

$$\vec{D} = \sum_j D_j \, \hat{e}_j \ , \tag{118}$$

we have the dot product

$$\vec{C} \cdot \vec{D} = \sum_i C_i \, D_i \ , \tag{119}$$

which, as promised, has the same form as (102) and uses the fact that the three unit vectors are orthonormal

$$e_i \cdot e_j = \delta_{ij} \ . \tag{120}$$

In three dimensions, we can introduce a new vector product, the *cross product*

$$\vec{C} \times \vec{D} = \sum_i \sum_j C_i \, D_j \, (\hat{e}_i \times \hat{e}_j) \ . \tag{121}$$

The cross product between orthonormal unit vectors is

$$e_i \times e_j = \sum_k \epsilon_{ijk} \, e_k \ , \tag{122}$$

where the Levi-Civita symbol is defined by [6]

$$\begin{aligned}\epsilon_{ijk} &= +1 \text{ for cyclic permutations of 123, like 312 etc.} \\ &= -1 \text{ for non-cyclic permutations of 123, like 132 etc.} \\ &= 0 \text{ if } i = j \text{ or } j = k \text{ or } i = k \ .\end{aligned} \tag{123}$$

Accordingly (121) simplifies to

$$\vec{C} \times \vec{D} = \sum_i \sum_j \sum_k \epsilon_{ijk}\, C_i\, D_j\, e_k \ , \tag{124}$$

Note that the cross product is perpendicular to both the $\hat{e}_1$ and $\hat{e}_2$ directions thanks to the properties of ϵ_{ijk}.

Exercise 23. Calculate the cross product of the two vectors

$$\vec{C} = 3\hat{e}_1 + 4\hat{e}_2 + 5\hat{e}_3 \quad ; \quad \vec{D} = \hat{e}_1 + \hat{e}_3 \ . \tag{125}$$

To define a vector in three dimensions, we need to examine how its components transform under rotations. There are three planes to consider and hence three rotation angles: θ_1 between the $x_1 - x_2$ axes, θ_2 between the $x_2 - x_3$ axes and θ_3 between the $x_3 - x_1$ axes. A general rotation in three dimensions is achieved using a rotation matrix $R(\theta_1, \theta_2, \theta_3)$ involving all three of these angles. Under a rotation, a three-vector transforms as

$$C'_i = \sum_j R_{ij}\, C_j \ , \tag{126}$$

again, identical in form to (107) but involving three equations (one for each value of i). Starting from (126), we write down the norm of a three-vector

$$\sum_i C'_i\, C'_i = \sum_j \sum_k R_{ij}\, R_{ik}\, C_j\, C_k \ , \tag{127}$$

where we have used k as the second dummy index. Since rotations leave the norm of a vector invariant we find that the rotation matrices must satisfy

$$R_{ij}\, R_{ik} = \delta_{jk} \ . \tag{128}$$

That way, when we impose this δ-function using the sum over k we obtain

$$\sum_i C'_i\, C'_i = \sum_j C_j\, C_j \ , \tag{129}$$

meaning that the norm is indeed invariant under a rotation of the axes. Notice that (128) is the statement that

$$R^T\, R = I \ , \tag{130}$$

[6] The non-vanishing symbols are $\epsilon_{123} = \epsilon_{312} = \epsilon_{231} = +1$; $\epsilon_{132} = \epsilon_{321} = \epsilon_{213} = -1$.

that the rotation matrices are orthogonal. Further taking the determinant on both sides we find

$$(\det R)^2 = 1 \ , \tag{131}$$

implying that

$$\det R = \pm 1 \ . \tag{132}$$

We can deduce the form of the matrix R from some examples. Consider a rotation where we keep the x_3 axis fixed. This would correspond to a rotation of the entire $x_1 - x_2$ plane about the stationary x_3 axis. Thus this rotation matrix would involve only a single angle, θ_1 (introduced on the previous page).

Physically, this rotation is exactly like a rotation in two dimensions, which also depends on a single angle (since the third dimension remains entirely unaltered). Thus the rotation matrix R must mirror the structure found in (9). So

$$R(\theta_1) \equiv \begin{bmatrix} \cos\theta_1 & \sin\theta_1 & 0 \\ -\sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \ , \tag{133}$$

reminiscent of the rotation matrix in (9).

Exercise 24. Invoke similar arguments to determine the complete structure of a general rotation matrix in three dimenions. Note that this should be a function of all three angles of rotation.

3.2 Vector Calculus

3.2.1 Gradient

Instead of the temperature of a floor (a two dimensional problem), we can now consider the temperature in the entire room $T(x_1, x_2, x_3)$, a scalar function of all three variables. An infinitesimal changes in T is governed by the relation

$$dT = \frac{\partial T}{\partial x_1} dx_1 + \frac{\partial T}{\partial x_2} dx_2 + \frac{\partial T}{\partial x_3} dx_3 = \sum_i \frac{\partial T}{\partial x_i}\, dx_i \ . \tag{134}$$

As before, we can recast this as a dot-product relation

$$dT = \overline{\nabla} T \cdot d\bar{l} \ , \tag{135}$$

where

$$\overline{\nabla} T = \sum_i \frac{\partial T}{\partial x_i} \hat{e}_i \ , \tag{136}$$

is the Gradient of T in three dimensions and

$$d\bar{l} = \sum_i dx_i\, \hat{e}_i \ , \tag{137}$$

is the length element in three dimensions. The *Gradient operator* is

$$\overline{\nabla} = \sum_i \hat{e}_i \frac{\partial}{\partial x_i} \ . \tag{138}$$

3.2.2 Divergence

Given an arbitrary vector in three dimensions

$$\vec{C} = \sum_k C_k \hat{e}_k \ , \tag{139}$$

the divergence of $\vec{C}$ is

$$\vec{\nabla} \cdot \vec{C} = \sum_i \sum_k \hat{e}_i \frac{\partial}{\partial x_i} \cdot C_k \hat{e}_k = \sum_i \sum_k \frac{\partial C_k}{\partial x_i} \delta_{ik} \ , \tag{140}$$

which simplifies to

$$\overline{\nabla} \cdot \overline{C} = \sum_i \frac{\partial C_i}{\partial x_i} \ . \tag{141}$$

Example 10. Calculate the divergence of the position vector

$$\overline{r} = x_1 \, \hat{e}_1 + x_2 \hat{e}_2 + x_3 \hat{e}_3 \ . \tag{142}$$

From (141)

$$\overline{\nabla} \cdot \overline{r} = \sum_i \frac{\partial r_i}{\partial x_i} = \frac{\partial r_1}{\partial x_1} + \frac{\partial r_2}{\partial x_2} + \frac{\partial r_3}{\partial x_3} \ . \tag{143}$$

r_1, r_2, r_3 refer to the three components of the position vector so

$$\overline{\nabla} \cdot \overline{r} = \frac{\partial x_1}{\partial x_1} + \frac{\partial x_2}{\partial x_2} + \frac{\partial x_3}{\partial x_3} = 3 \ . \tag{144}$$

3.2.3 Curl

The cross product, as before, allows us to introduce the *curl* of a vector

$$\overline{\nabla} \times \vec{C} = \sum_i \sum_j \sum_k \epsilon_{ijk} \, \partial_i \, C_j \, \hat{e}_k \ , \tag{145}$$

where we have introduced the notation

$$\partial_i \equiv \frac{\partial}{\partial \, x_i} \ . \tag{146}$$

Example 11. Compute the curl of the vector $\vec{C} = -x_2\,\hat{e}_1 + x_1\,\hat{e}_2$.

Expand the sum over k in (145) to obtain

$$\begin{aligned}\overline{\nabla}\,\times\,\vec{C} \;&= \sum_i \sum_j \epsilon_{ij1}\,\partial_i\,C_j\,\hat{e}_1 + \sum_i \sum_j \epsilon_{ij2}\,\partial_i\,C_j\,\hat{e}_2 + \sum_i \sum_j \epsilon_{ij3}\,\partial_i\,C_j\,\hat{e}_3\ , \\ &= (\epsilon_{231}\,\partial_2\,C_3 + \epsilon_{321}\,\partial_3\,C_2)\,\hat{e}_1 + (\epsilon_{312}\,\partial_3\,C_1 + \epsilon_{132}\,\partial_1\,C_3)\,\hat{e}_2\ , \\ &\quad +(\epsilon_{123}\,\partial_1\,C_2 + \epsilon_{213}\,\partial_2\,C_1)\,\hat{e}_3\ , \\ &= (\partial_2\,C_3 - \partial_3\,C_2)\,\hat{e}_1 + (\partial_3\,C_1 - \partial_1\,C_3)\,\hat{e}_2 + (\partial_1\,C_2 - \partial_2\,C_1)\,\hat{e}_3\ . \end{aligned} \tag{147}$$

We have expanded the sums over i and j in the second and third lines above and we have used the properties of the Levi-Civita symbol in the third line. Substituting for the various components, we find

$$\begin{aligned}\overline{\nabla}\,\times\,\vec{C} &= (\partial_2\,\{0\} - \partial_3\,\{x_1\})\,\hat{e}_1 + (\partial_3\,\{-x_2\} - \partial_1\,\{0\})\,\hat{e}_2 + (\partial_1\,\{x_1\} - \partial_2\{-x_2\})\,\hat{e}_3\ , \\ &= 2\,\hat{e}_3 \end{aligned} \tag{148}$$

Exercise 25. Sketch the vector field $\vec{C}$ from the example above. From the figure, convince yourself that the "curl of a vector" is a suitable name.

3.3 Some results in index notation

We now derive a few results that further illustrate the power of index notation. Results that often take a page to derive may be obtained in a few lines.

A 'repeated index' is one which appears twice on the same side of the 'equal to' sign. We now introduce the Einstein summation convention which tells us that any repeated index is always summed over, even if the $\sum$ symbol is not explicitly written. So

$$a_j\,b_j \;\text{ is completely equivalent to }\; \sum_j a_j\,b_j\ , \tag{149}$$

even though the first expression does not have an explicit summation sign.

We now compute some common expressions, starting with the curl of a gradient.

$$\overline{\nabla}\,\times\,\overline{\nabla}\,\Phi\ , \tag{150}$$

where Φ is some scalar function. Using (138) and (145) we find

$$\overline{\nabla}\,\times\,\overline{\nabla}\,\Phi = \hat{e}_i\partial_i\,\times\,\hat{e}_j\,\partial_j\,\Phi\ . \tag{151}$$

Since the unit vectors are the only vector quantitiies present, we have

$$\overline{\nabla}\,\times\,\overline{\nabla}\,\Phi = \partial_i\,\partial_j\,\Phi\,(\,\hat{e}_i \times \hat{e}_j) = \epsilon_{ijk}\,\partial_i\,\partial_j\,\Phi\,\hat{e}_k\ . \tag{152}$$

We simply note that

$$\epsilon_{ijk}\,\partial_i\,\partial_j\,\Phi\,\hat{e}_k = 0\ , \tag{153}$$

because ϵ_{ijk} is antisymmetric, by definition, in i and j while the expression $\partial_i\,\partial_j$ is symmetric in the *same two indices*. Thus

$$\overline{\nabla}\ \times\ \overline{\nabla}\,\Phi = 0\ . \tag{154}$$

Exercise 26. Verify explicitly equation (153) which teaches us that the product of an antisymmetric object with a symmetric one always vanishes.

Exercise 27. Prove, using index notation, that the divergence of a curl (of any vector field) always vanishes.

Exercise 28. Prove that

$$\epsilon_{mil}\,\epsilon_{jkl} = \delta_{mj}\delta_{ik} - \delta_{mk}\delta_{ij}\ . \tag{155}$$

Let us move on to a slightly more complicated computation. Consider the curl of a curl.

$$\overline{\nabla}\times(\overline{\nabla}\times\vec{A}) = \hat{e}_i\,\partial_i\ \times\ \epsilon_{jkl}\,\partial_j\,A_k\,\hat{e}_l = \epsilon_{mil}\epsilon_{jkl}\,\partial_i\partial_j A_k\,\hat{e}_m\ . \tag{156}$$

Using the result of **Exercise 28.** we find that (156) simplifies to

$$\overline{\nabla}\times(\overline{\nabla}\times\vec{A}) = \partial_i\partial_j A_i\,\hat{e}_j\ -\ \partial_i\partial_i A_k\,\hat{e}_k\ , \tag{157}$$

which is the final result and may be written in the more familiar form

$$\overline{\nabla}\times(\overline{\nabla}\times\vec{A}) = \overline{\nabla}(\overline{\nabla}\cdot\vec{A})\ -\ \overline{\nabla}^2\,\vec{A}\ . \tag{158}$$

Exercise 29. Verify that (158) follows from (157).

3.4 Group Theory

3.4.1 The SO(3) Group

We now return to (126) and our study of rotation matrices in three dimensions. As in two dimensions, the three-dimensional rotation matrices form a group under multiplication. With three angles involved, the group is now called $SO(3)$ with the S (Special) and O (Orthogonal) indicative of the same properties as in two dimensions.

We introduce the following simple definition of the group $SO(3)$

$$SO(3) = \{R \in M_3(\mathcal{R}), \det R = 1, R^T\,R = 1\}\ , \tag{159}$$